WINDOWS

Quick Python and VS Code setup

Follow this page on Windows, or page 2 on Mac. Then finish on page 3. Allow about 20–30 minutes, plus download time.

1 Install Python from the Microsoft Store

Open the **Microsoft Store**. Search for **Python Install Manager** and install the one published by the **Python Software Foundation**. Open **Windows PowerShell** from the Start menu. Copy and paste this block, press Enter, and wait for it to finish:

```
pymanager install 3.13
py -3.13 --version
```

You should see `Python 3.13.x`. Close PowerShell when finished.

2 Install VS Code

Download and install Visual Studio Code. Open it, click **Extensions** in the left sidebar, and install these two extensions by **Microsoft**: **Python** and **Jupyter**.

3 Create a project folder

Create a folder named `RTS-Research` in Documents. In VS Code, choose **File > Open Folder** and open that folder. Choose **Terminal > New Terminal**. Select **PowerShell** from the terminal dropdown if needed.

4 Install all libraries at once

Copy and paste this whole block into that PowerShell terminal. Press Enter if the last line is waiting. Leave it open and wait for installation to finish:

```
py -3.13 -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install numpy pandas scipy `
  matplotlib seaborn scikit-learn openpyxl ipykernel notebook
.\.venv\Scripts\python.exe -m pip check
```

Done when: the last line says `No broken requirements found`. This installs tools for data cleaning, Excel, plotting, ML, and Jupyter. The `.venv` folder keeps your project libraries together; no activation is needed.

Next: skip the Mac page and go to page 3.

MAC

Quick Python and VS Code setup

Windows users: skip this page.

1 Install Python

Open the Python 3.13.16 download page. Under **Files**, download the **macOS installer**. Open the downloaded `.pkg` file and follow the installation steps.

In Finder, open **Applications > Python 3.13** and double-click **Install Certificates.command**. Let it finish.

2 Install VS Code

Download Visual Studio Code for Mac. Unzip it if needed, drag the app into **Applications**, and open it. Click **Extensions** in the sidebar and install these two extensions by **Microsoft**: **Python** and **Jupyter**.

3 Create a project folder

Create a folder named `RTS-Research` in Documents. In VS Code, choose **File > Open Folder** and open that folder. Then choose **Terminal > New Terminal**.

4 Install all libraries at once

Copy and paste this whole block into the terminal. Press Enter if the last line is waiting. Leave it open and wait for installation to finish:

```
python3.13 -m venv .venv
./.venv/bin/python -m pip install --upgrade pip
./.venv/bin/python -m pip install numpy pandas scipy \
  matplotlib seaborn scikit-learn openpyxl ipykernel notebook
./.venv/bin/python -m pip check
```

Done when: the last line says `No broken requirements found`. This installs tools for data cleaning, Excel, plotting, ML, and Jupyter. The `.venv` folder keeps your project libraries together; no activation is needed.

Next: go to page 3.

EVERYONE

Open a notebook and check your setup

5 Create your first notebook

1. Open the Command Palette with **Ctrl+Shift+P** (Windows) or **Cmd+Shift+P** (Mac). Search for **Create: New Jupyter Notebook**.
2. Save it as `setup_check.ipynb` in `RTS-Research`.
3. Click **Select Kernel** at the top right, choose **Python Environments**, and select the project's `.venv`.
4. Paste this into a **code cell** and press **Shift+Enter**:

```
import sys
import numpy as np
import pandas as pd
import scipy, seaborn, sklearn, openpyxl
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

print("Python:", sys.version.split()[0])
print("Environment:", sys.executable)
model = LinearRegression().fit([[1], [2], [3]], [2, 4, 6])
print("Test prediction:", round(model.predict([[5]])[0], 1))
plt.plot([1, 2, 3], [2, 4, 6], marker="o")
plt.show()
```

Done when: you see `Test prediction: 10.0`, a line plot, and an environment path containing `.venv`. Save your notebook. You are ready!

If something goes wrong

Windows cannot find pymanager or py: Close and reopen VS Code and PowerShell after installing Python Install Manager. If `py` still points to an older installation, ask your leader to check the Python launcher.

Cannot find .venv: Open your `RTS-Research` folder in VS Code, open a new terminal, and rerun step 4. If a command fails, fix that error before continuing.

Notebook says a module is missing: Select the project's `.venv` kernel again. If needed, rerun the library installation line.

Still stuck after 10 minutes? Send your group leader a screenshot of the full error and the command you ran.

Official help: Python on Windows | Jupyter in VS Code. Next time, open the same folder and notebook; you do not need to reinstall.